

BRAC University
ECO101: Introduction to Microeconomics
Practice Problems Set 1
Spring 2025

Reminder:

- Solve the problems provided in the slides and solved in class for a better understanding of the following problems.
- Do not forget to solve the problems provided in Quiz 1 and Assignment 1 for the problems under the first two topics.
- Do not use AI to solve the problems, as it often gives incorrect answers.

Elasticity:

1. The equation for a supply curve is $P = 3Q_s - 8$. What is the elasticity in moving from a price of 4 to a price of 7? State whether the supply is elastic, inelastic, or unit elastic.

2. Muse, the local art museum, is considering raising its admission fee by 10 percent. Muse last raised their fee three years ago. You have been asked to project whether the new admission fee increase will lead to an increase or decrease in revenues. For the analysis, you have been given data pertaining to the previous fee increase:

	Before increase	After increase
Visitors per day:	2,000	1,400
Average price paid:	\$8.00	\$10.00

a) Based on the figures provided, calculate the price elasticity of demand for visits to Muse. b) Based on your calculation above, would you expect revenues to increase or decrease if tolls increase by another 10 percent? Why?

3. Use the table to answer the following questions, which show the changes in prices and quantities of coffee and tea between May and August.

	March		April	
Product	Price (per unit)	Quantity	Price (per unit)	Quantity
Coffee	120	1300	150	1550
Tea	80	1850	100	2150

- a) Calculate the Price Elasticity of Supply (PES) for coffee. Comment on the PES.
- b) Calculate the Cross Price Elasticity of Demand for coffee (XED) due to the price change in tea. Comment on the XED.
- c) Given that the initial average income level was BDT 30000 in March. But the income level in the economy increased by 20% in April. What is the income elasticity of demand for tea given the change in the income level? Comment on the YED.
- d) After April, if the government imposes a policy that reduces the overall income level in the economy, what do you think will happen to the demand for tea?

Efficiency - Tax:

4. Consider the market for cigarettes, which has a steeper downward-sloping demand curve than the usual upward-sloping supply curve. The market equilibrium price is Tk 8 per pack, and the equilibrium quantity is 100 million packs.

Now, to curb smoking habits, the government imposes an excise tax on cigarettes, where the producer must pay a tax of Tk 4 per pack. After tax, the supply curve shifts upward, and the consumers now buy 70 million packs, and they pay a price of Tk 10 per pack. Using an appropriate diagram, answer the following questions:

- a) Show the above tax situation in a properly labeled diagram, marking the areas for government tax revenue and the deadweight loss.
- b) Calculate the total tax revenue.
- c) Calculate the deadweight loss.
- d) Calculate the loss in producer's surplus due to tax.

5. Consider the market for Good X.

Demand: $Q_d = 80 - 5P$ and Supply: $Q_s = 5P - 20$

- a) Given the information, find the market equilibrium price and quantity for good X. b) Calculate the consumer surplus, the producer surplus, and the total surplus for good X. Suppose the government decides to impose a tax of TK 5 per unit on good X. c) Given the tax, find the new price that buyers will pay and the new price that sellers will receive.
- d) Find the new CS, PS and TS.
- e) Calculate the tax revenue generated by the government and the deadweight loss from the tax.

6. The health ministry is evaluating the data of the soft beverages market.

Demand function: $Q_d = 290 - 5P$ Supply function: $Q_s = 5P - 60$

- a) Find the total surplus.

The authority has decided to impose a tax of Tk. 10 per unit on soft drinks.

- b) What is the government tax revenue?
- c) What is the deadweight loss after tax?